Federated Optimization in Heterogeneous Sensor Networks

ABSTRACT: We investigate asynchronous federated learning across edge devices.

METHODOLOGY: Decentralized stochastic gradient descent with momentum.

RESULTS: Empirical evaluation demonstrates 42 percent communication reduction.

LIMITATIONS: Edge straggler latencies remain unaddressed in high jitter channels.

FUTURE WORK: Investigate adaptive deadline clustering and straggler drop policies.